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Constraint equation $\rightarrow I_x = i_a + i_b$

$$i_c = 40 I_x$$

KVL for loop-1

$$-20 + (i_a - i_c) + 25(i_a + i_b) + 4i_a = 0$$

$$\rightarrow 30i_a + 25i_b - i_c = 20$$

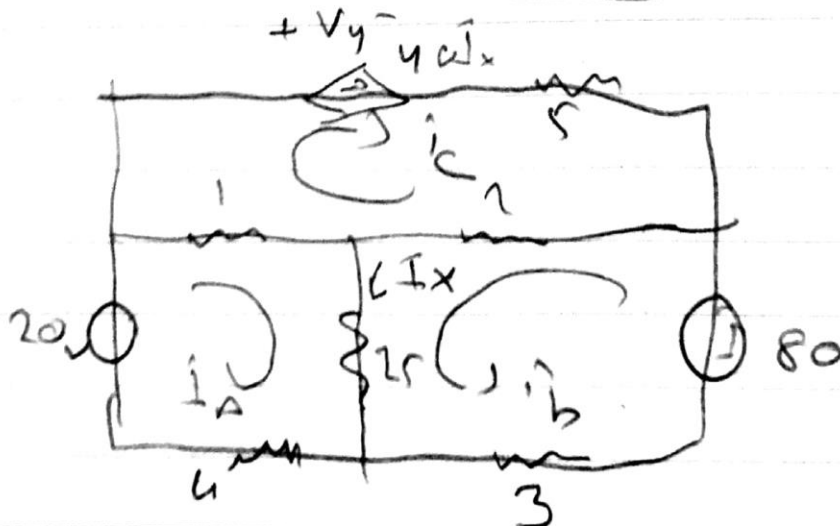
KVL for loop-2

$$2(i_b + i_c) + 25(i_a + i_b) + 3i_b - 80 = 0$$

$$\rightarrow 25i_a + 30i_b + 2i_c = 80$$

KVL for loop-3

$$(i_c - i_a) + V_y + 5i_c + 2(i_c + i_b) = 0$$



Tolga konat
200555041

1 → Node Voltage Method

$$V_1 = 180 \quad V_3 = 108$$

$\frac{V_2}{10}$

kcl → node 2

$$\frac{V_0 - 180}{50} + \frac{V_0}{15} + \frac{V_0 - 108}{30} = 0$$

$$\frac{3V_0 - 3 \cdot 60 \cdot 3}{150} + \frac{10V_0}{150} + \frac{5V_0 - 5 \cdot 36 \cdot 3}{150} = 0 \quad \boxed{V_0 = 60}$$

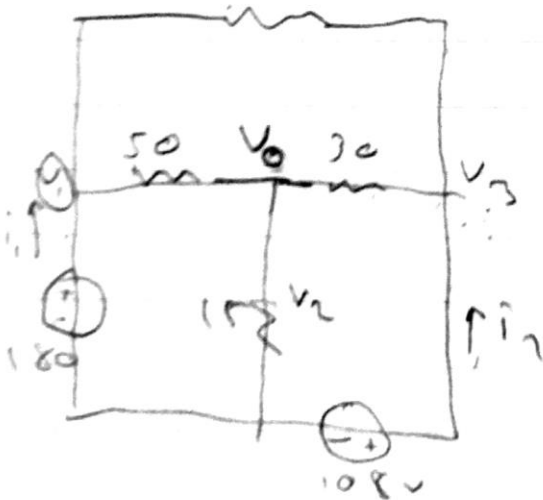
$$-i_1 + \frac{V_1 - V_3}{90} + \frac{V_1 - V_2}{30} = 0$$

$$-i_1 + \frac{V_1 - 180}{90} + \frac{V_1 - V_2}{30} = 0 \Rightarrow -i_1 + \frac{180 - 108}{90} + \frac{180 - 60}{30} = 0$$

$$i_1 = \frac{180 - 108}{90} + \frac{180 - 60}{30}$$

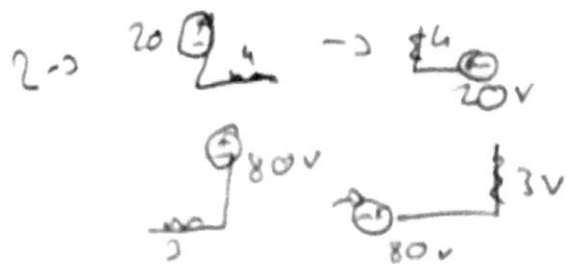
$$i_1 = \frac{4}{5} + \frac{12}{5} = \frac{16}{5} \text{ A} \Rightarrow i_1$$

$$i_2 = i_1 + \frac{4}{5} \cdot \frac{8}{6} \quad \boxed{i_2 = \frac{4}{5} \text{ A}}$$



$$i_1 \rightarrow \frac{16}{5} \text{ A}$$

$$i_2 \rightarrow \frac{4}{5} \text{ A}$$



constraint eq

$$\rightarrow i_x = \frac{V_1 - 0}{20}$$

KCL for node-1

KCL for node-1

$$\frac{V_1 - 20}{4} + \frac{V_1 - 0}{1} + \frac{40i_x}{4} = 0 \rightarrow \boxed{25V_1 + 12V_2 = 100}$$

KCL for node-2

$$\frac{V_2 - V_1}{1} + \frac{V_2}{20} + \frac{V_2 - V_3}{2/20} = \boxed{-50V_1 + 7V_2 - 25V_3 = 0}$$

KCL for node-3

$$\frac{V_3 - V_1}{2} + \frac{V_3 - 0}{3} - 40i_x = \boxed{-63V_1 + 25V_3 = 800}$$

